

EMCH 260

SOLID MECHANICS

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Strain Transformation

Chapter 10.3

- ✓ Define stress-strain relationship
- ✓ Material property relationships

GENERALIZED HOOKE'S LAW

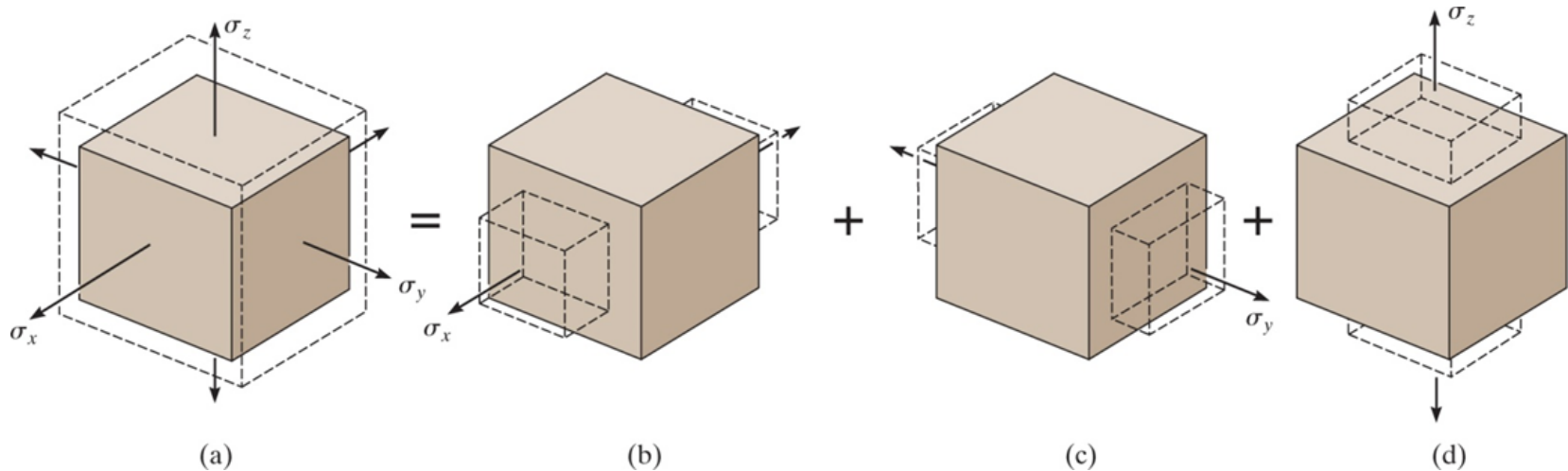
- Material at a point subjected to a state of **triaxial stress**, with associated strains.
- We use **principle of superposition**, Poisson's ratio ($\varepsilon_{\text{lat}} = -\nu\varepsilon_{\text{long}}$), and Hooke's law ($\varepsilon = \sigma E$) to relate stresses to strains, in the uniaxial direction.
- With σ_x applied, element elongates in the x direction and strain in this direction is

$$\varepsilon'_x = \frac{\sigma_x}{E}$$

GENERALIZED HOOKE'S LAW, CONT'D

- With σ_y applied, element contracts with a strain ε''_x in the x direction,
$$\varepsilon''_x = -\nu \frac{\sigma_y}{E}$$

- Likewise, With σ_z applied, a contraction is caused in the z direction,
$$\varepsilon'''_x = -\nu \frac{\sigma_z}{E}$$



GENERALIZED HOOKE'S LAW, CONT'D

- By using the principle of superposition,

$$\begin{aligned}\varepsilon_x &= \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] \\ \varepsilon_y &= \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)] \\ \varepsilon_z &= \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]\end{aligned}\quad (10-18)$$

MATERIAL-PROPERTY RELATIONSHIPS

Generalized Hooke's law

- If we apply a shear stress τ_{xy} to the element, experimental observations show that it will deform only due to shear strain γ_{xy} . Similarly for τ_{xz} and γ_{xz} , τ_{yz} and γ_{yz} . Thus, Hooke's law for shear stress and shear strain is written as

$$\gamma_{xy} = \frac{1}{G} \tau_{xy} \quad \gamma_{yz} = \frac{1}{G} \tau_{yz} \quad \gamma_{xz} = \frac{1}{G} \tau_{xz} \quad (10-19)$$

MATERIAL-PROPERTY RELATIONSHIPS, CONT'D

Relationship involving E , ν , and G

- We stated in Ch.3 (textbook page 108):

$$G = \frac{E}{2(1 + \nu)} \quad (10 - 20)$$

- Relate principal strain to shear stress,

$$\varepsilon_{\max} = \frac{\tau_{xy}}{E} (1 + \nu) \quad (10 - 21)$$

- Note that since $\sigma_x = \sigma_y = \sigma_z = 0$, then from Eq. 10-18, $\varepsilon_x = \varepsilon_y = 0$. Substitute into transformation Eq. 10-19,

$$\varepsilon_1 = \varepsilon_{\max} = \frac{\gamma_{xy}}{2}$$

MATERIAL-PROPERTY RELATIONSHIPS, CONT'D

Relationship involving E , ν , and G

- By Hooke's law, $\gamma_{xy} = \tau_{xy}/G$. So $\epsilon_{max} = \tau_{xy}/2G$.
- Substitute into Eq. 10-21 and rearrange to obtain Eq. 10-20.

Dilatation and Bulk Modulus

- Consider a volume element subjected to principal stresses σ_x , σ_y , σ_z .
- Sides of element are dx , dy and dz , and after stress application, they become $(1 + \epsilon_x)dx$, $(1 + \epsilon_y)dy$, $(1 + \epsilon_z)dz$, respectively.

MATERIAL-PROPERTY RELATIONSHIPS, CONT'D

Dilatation and Bulk Modulus

- Change in volume of element is

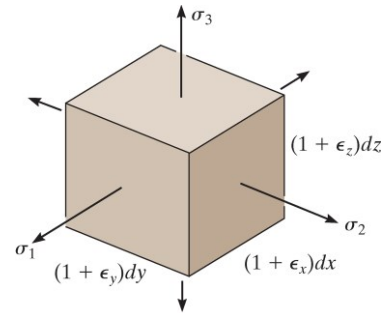
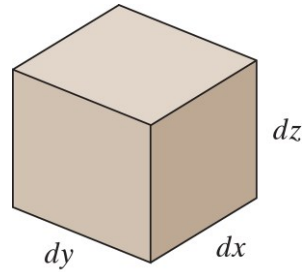
$$\delta V = (1 + \varepsilon_x)(1 + \varepsilon_y)(1 + \varepsilon_z)dx dy dz - dx dy dz$$

- Change in volume per unit volume is the “volumetric strain” or dilatation e .

$$e = \frac{\delta V}{dV} = \varepsilon_x + \varepsilon_y + \varepsilon_z \quad (10-22)$$

- Using generalized Hooke’s law, we write the dilatation in terms of applied stress.

$$e = \frac{1 - 2\nu}{E}(\sigma_x + \sigma_y + \sigma_z) \quad (10-23)$$



MATERIAL-PROPERTY RELATIONSHIPS, CONT'D

Dilatation and Bulk Modulus

- When volume element of material is subjected to uniform pressure p of a liquid, pressure is the same in all directions.
- As shear resistance of a liquid is zero, we can ignore shear stresses.
- Thus, an element of the body is subjected to principal stresses $\sigma_x = \sigma_y = \sigma_z = -p$. Substituting into Eq. 10-23 and rearranging,

$$\frac{p}{e} = \frac{E}{3(1 - 2\nu)} \quad (10 - 24)$$

MATERIAL-PROPERTY RELATIONSHIPS, CONT'D

Dilatation and Bulk Modulus

- This ratio (p/e) is similar to the ratio of linear-elastic stress to strain, thus terms on the RHS are called the **volume modulus of elasticity** or the **bulk modulus**. Having same units as stress with

symbol k ,

$$k = \frac{E}{3(1 - 2\nu)} \quad (10 - 25)$$

- For most metals, $\nu \approx 1/3$ so $k \approx E$.
- From Eq. 10-25, theoretical maximum value of Poisson's ratio is therefore $\nu = 0.5$.
- When plastic yielding occurs, $\nu = 0.5$ is used.

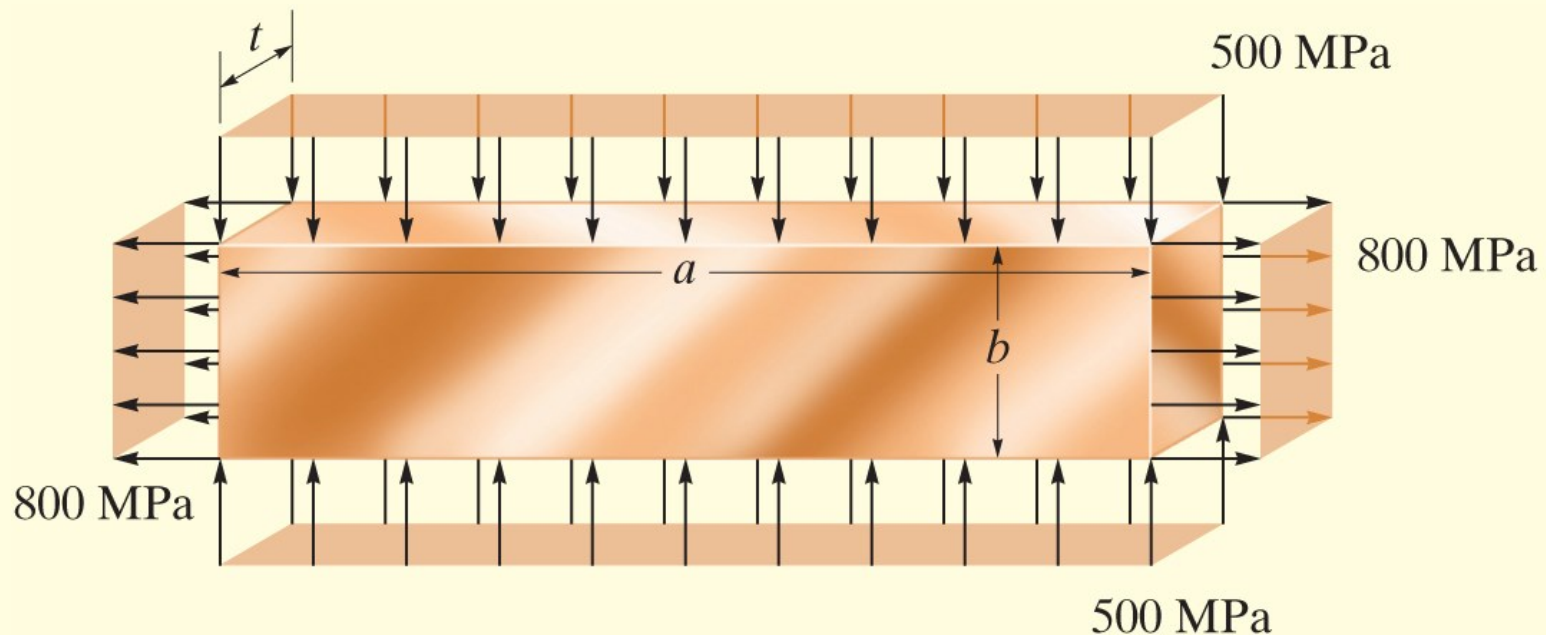
IMPORTANT NOTES

IMPORTANT

- When homogeneous and isotropic material is subjected to a state of triaxial stress, the strain in one of the stress directions is influenced by the strains produced by all stresses. This is the result of the Poisson effect, and results in the form of a generalized Hooke's law.
- A shear stress applied to homogeneous and isotropic material will only produce shear strain in the same plane.
- Material constants, E , G and ν are related mathematically.
- Dilatation, or volumetric strain, is caused only by normal strain, not shear strain.
- The bulk modulus is a measure of the stiffness of a volume of material. This material property provides an upper limit to Poisson's ratio of $\nu = 0.5$, which remains at this value while plastic yielding occurs.

IN-CLASS EXAMPLE

The copper bar is subjected to a uniform loading along its edges as shown. If it has a length $a = 300$ mm, $b = 500$ mm, and $t = 20$ mm before the load is applied, determine its new length, width, and thickness after application of the load. Take Young's modulus of copper as 120 GPa and the Poisson's ratio as 0.34.



IN-CLASS EXAMPLE, cont'd

Solutions

- From the loading we have

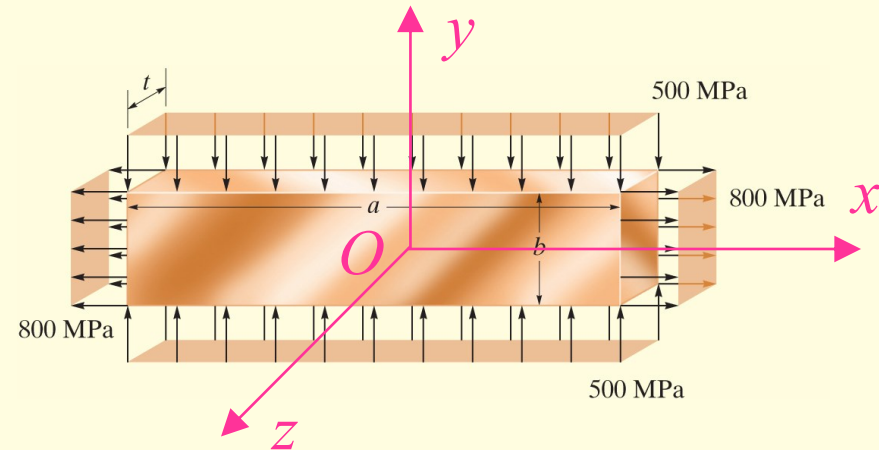
$$\sigma_x = 800 \text{ MPa}, \sigma_y = -500 \text{ MPa}, \sigma_z = 0$$

- The associated normal strains are determined from the generalized Hooke's law,

$$\varepsilon_x = \frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z) = 0.00808$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_x + \sigma_z) = -0.00643$$

$$\varepsilon_z = \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y) = -0.000850$$



- The new bar length, width, and thickness are therefore

$$a' = 300 + 0.00808(300) = 302.4 \text{ mm (Ans)}$$

$$b' = 50 + (-0.00643)(50) = 49.68 \text{ mm (Ans)}$$

$$t' = 20 + (-0.000850)(20) = 19.98 \text{ mm (Ans)}$$

RELEVANT PRACTICE

- **Assignment 4 on Blackboard, problem #3-4 (go to Blackboard under the “Assignments” folder, **Assignment 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**
- **Optional Extra Exercise 4 on Blackboard, problem #3-4 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 516 – 523**
 - **In particular, Examples 10.9 – 10.11**